MICROSCOPY

I. OBJECTIVES

- To demonstrate skill in the proper utilization of a light microscope.
- To demonstrate skill in the use of ocular and stage micrometers for measurements of cell size.
- To recognize and describe differences in cell morphology and determine cell size at all magnifications.

II. INTRODUCTION

There are several types of microscope (simple, compound, light or bright-field, dark-field, electron, fluorescence, interference, etc.) but the one most commonly used for bacteriological purposes is the bright-field or light microscope. This microscope is composed of a light source, a substage condenser, a stage where the specimen slide is placed, 3-4 objective lenses and ocular lenses. These parts are attached together by a solid backbone or spine. The light from a lamp passes through the condenser onto the specimen such that the field of observation becomes bright. The image of the specimen then passes through a series of lenses to the eye of the observer. The lens series closer to the object or specimen is called the objective and the one close to the eye of the observer is called the ocular lens.

Parts of Microscope

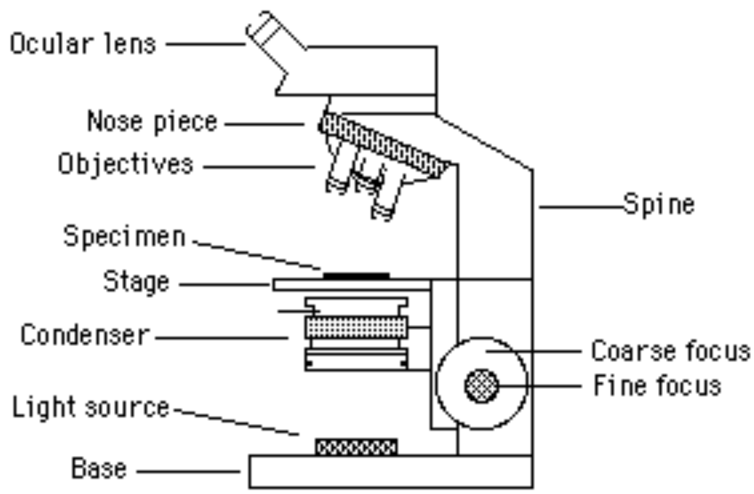
The light that illuminates the specimen in a bright field microscope comes from a small lamp and is focused by a unit called the condenser that contains a lens and a series of filters. The condenser can be lowered or raised and contains an iris diaphragm that can be adjusted manually. The size of the iris aperture (opening) determines the amount of light reaching the specimen. In general, with higher lens magnifications, more light is needed to observe the specimen properly. Too little or too much light causes decreased visibility of the specimen. Another way to control the amount of illumination is by changing the vertical position of the condenser. When the condenser is lowered, less light reaches the specimen and vice versa.

The resolving power is defined as the ability of the microscope to distinguish between two individual adjacent points. We can say the resolving power of microscope A is less than that of microscope B if looking through microscope A at a certain specimen, you see a single point while looking at the same specimen through microscope B, you see two separate points.

Dr. Eby Bassiri ebassiri@sas.upenn.edu

Both the lens system and the light quality affect the resolving power of the microscope. Sometimes it is necessary to place different filters (e.g., blue filter) on the condenser to absorb the longer wavelengths of the visible spectrum and increase the resolution. The maximum resolution of the light microscope is 0.2 μm. Most bacteria range in size from about 1 μ to 10 μ.

Note: 1 inch = 2.54 cm; 1 cm = 10 mm; 1 mm = 1000 μm or simply μ (micron)



The objective lenses are located on a revolving nosepiece. Most light microscopes contain objectives that magnify the specimen 4X (4 times), 10X, 40X and 100X. The 40X is called high dry lens and the 100X is called the oil immersion lens. Newer light microscopes are parfocal, meaning that if you focus on a specimen with any of the objectives and then change to another objective, the specimen would still stay in focus, except for minor fine adjustment.

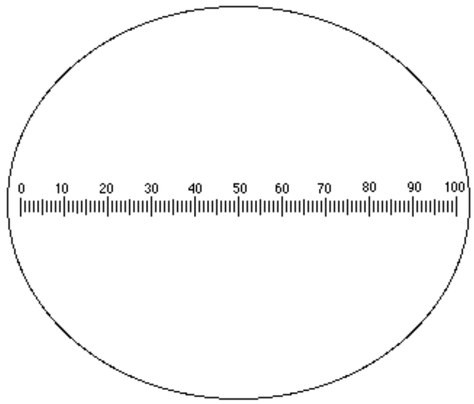
The stage separates the condenser from objective lenses and is usually in the shape of a solid metallic square or rectangle. The specimen is placed on the stage and is held in position by two movable arms. The stage can be raised or lowered along the spine to change the distance between the specimen and the objective and thus bring the image into focus.

The ocular lens is a part of the eyepiece. Most ocular lenses magnify the image another 10X. The final magnification of the specimen is obtained by multiplying the objective by the ocular powers. Thus, a light microscope can magnify an object 40 to 1,000 times.

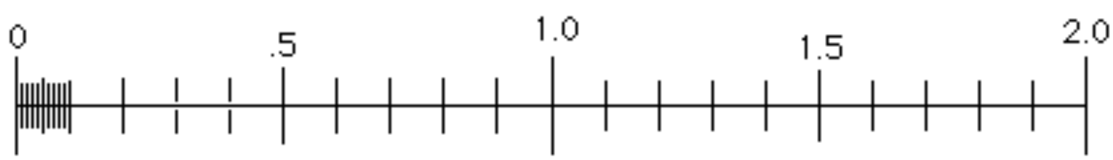
Dr. Eby Bassiri ebassiri@sas.upenn.edu

Bacterial Size Measurements

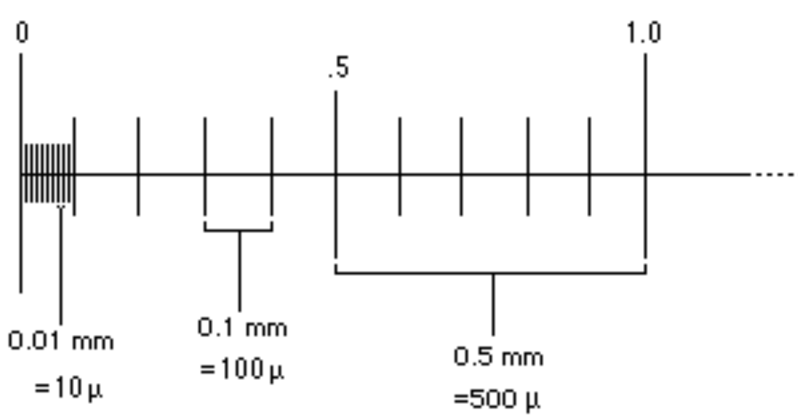
To be able to measure the size of microorganisms, an ocular micrometer disc is placed in one of the oculars. The disc has numbered lines on it as is shown below:



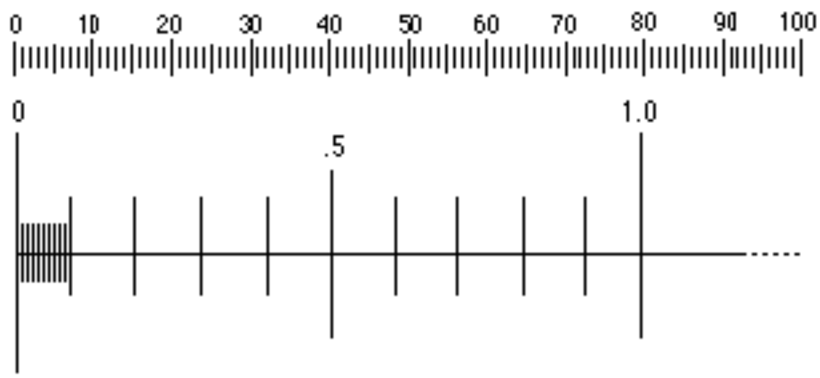
The units might be different on different ocular micrometers; i.e., some go up to 10 or 50 while others go up to 100. Our main concern here is to determine the length of one unit of the ocular micrometer. For this purpose, we need to calibrate this unit against a known length. There are prepared slides on the market called stage micrometers which have a scale of known length etched in the glass. The scale when observed under the microscope looks as below:



The total given length for the above is 2.0 mm (only 1.0 mm is shown above) as measured by the manufacturer. Thus it can be seen that the units would be as follows:



Our next step is to superimpose the ocular and stage micrometers together and measure the length of one unit of the ocular based on the given length of the stage micrometer. As an example, suppose that for a certain microscope, after lining up the micrometers, we get the following image under the 10X objective:



This shows that 80 ocular units are equal to 1.0 mm (1000 microns) for this particular microscope and objective lens. This corresponds to each ocular unit being 12.5 μ long at this specific magnification (10X).

The ocular micrometer should be calibrated for each objective lens and recorded. This is necessary only once for each magnification and each microscope. After such calibrations, microbial size can be measured directly from the ocular micrometer.

Transportation of Microscope

Special care should be exercised in moving the microscope to avoid dropping or bumping into other objects. The microscope should be lifted for transport by its spine and one hand should always be placed under the base. Carry the microscope in a vertical position. When not in use, always put the cover on to protect it from dust.

Cleaning of Microscope

Always clean your microscope before and after use. Lower the stage to its lowest position. Remove slides left on the stage, if any, and look for oil or dust on the stage and the lenses. Crumple a piece of lens paper and clean the lenses (both oculars and objectives) and the stage, using circular motion. Repeat a few times until all are clean. Only lens paper should be used as other types of paper (e.g. bibulous paper) may scratch the surface of the delicate lenses. If you have used prepared slides, they should also be cleaned of any oil in the same way. If the oil cannot be removed from the lenses or the slides, a piece of lens paper dampened with methanol should be used first, followed by dry lens paper.

III. LABORATORY SUPPLIES

Bright field microscope	1/student
Stage micrometer	2/table
Prepared slides set #1	1/table
Lens paper	1/table
Immersion oil	1/table
Methanol bottle	1/table

IV. PROCEDURE (Each student works independently for this exercise)

1. Get your assigned microscope, record its number and compare its parts to the drawing and learn the function of each part.

2. Make sure that all the lenses (ocular as well as objectives), stage and other parts of your microscope are clean. If not, clean them with lens paper or methanol as described previously.

3. Plug in the microscope and turn on the light switch. Look through the ocular and make sure that the ocular micrometer is in place. Obtain a stage micrometer and place it on the stage in between the mechanical arms.

4. Move the condenser to its lowest position and open the iris diaphragm 3/4 of the way. Turn the light source to mid position. While looking at the stage, swing the 4X lens into position and bring up the stage to its highest position. Turn the coarse focus knob while looking through the ocular until the image of the stage micrometer appears sharply. You may try the fine adjustment knob to fine-tune the focusing. Calibrate your 4X objective as explained before. Record your calculations. Repeat this above procedure with the 10X and 40X objectives.

5. Next, place <u>a very small drop</u> of immersion oil on the slide and swing the 100X objective gently into position. At this point, the objective and the specimen slide are connected to each other by the oil droplet. For oil immersion microscopy, raise the condenser to its highest position and open the iris diaphragm completely. Also slide the light source switch to maximum. Record your calibration of this lens.

6. Remove the stage micrometer, clean it with lens paper (and with methanol, if needed) and replace it with a prepared slide of colored threads or letters. Observe these under 4X, 10X and 40X. Next try the prepared slides of bacterial types. There are smears of 3 different bacterial shapes on each slide. Again start with the 4X objective and increase the power up to the oil immersion lens. Measure the width and length of microbial cells (or diameter in case of round cells) at this highest magnification in these last three smears. Make a sketch of each bacterial shape and report its size in microns on the drawings.

Results of the Microscopy Lab Exercise

NAME ________________________________DATE ____________ GROUP NAME ________

The number of your assigned microscope:
Length of one ocular division:

40X ___________ 100X ___________ 400X __________ 1000X ___________

==

Organism Shape	Drawing	Size* Average Length	Size* Average Width

*If the bacterium is round in shape (coccus), report the diameter in both length and height columns.

Dr. Eby Bassiri ebassiri@sas.upenn.edu